Google Reviews LocalLens (Beginner): Led by Dhruv Sehgal

Local Lens

1. Project Summary

- **Project Name:** Local Lens
- **Team Members:** Aiden Hernandez, Karsten Lowe, Aurora Cong, Ngo Nguyen Anh Tran
- **Date:** March 6, 2026
- **Status:** Complete

2. Goals & Objectives

This project aims to use business metadata and Google reviews to provide services on:

- Where businesses can open new branches or their first branches
- How to help customers choose the best company based on their needs

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3. Data Infrastructure & Pipeline

This project utilizes business metadata and review datasets for each state in the U.S.A. We also use external datasets for determining cities within states and combining redundant business categories into a single category to make creating the dashboard more manageable. These datasets are cleaned and processed before being merged into a global dataset for Tableau.
We are moving data through four distinct stages:

1. **Ingestion:** Loading a JSON file directly crashes standard memory. In order to overcome this, we used custom `parse()` generators . This allows us to filter for specific columns—like `gmap_id`, `rating`, and `text`—without loading the entire metadata object.
2. **Joining datasets:**
   - The city dataset was joined to the business dataset by matching each business to the nearest city using latitude and longitude coordinates.
   - Additionally, a category mapping table was used to join broad categories to each business category.
   - The business metadata and the review datasets were merged on `gmap_id`.
3. **Processing Tools:** `Pandas`, `Scikit-Learn (StandardScaler,BallTree, tfidfVectorizer)`.
   a. **Sentiment Analysis:** Used `VADER` to turn review sentences into a "Compound Score" (e.g., "The food was great!" becomes `0.85`)

b. **Reviews Analysis:** Used `TfidfVectorizer` to calculate TF-IDF scores of all bigrams (e.g. “good food” or “customer service”) in reviews. Higher scores indicate more relative importance of a word in relation to all reviews (in a specific region classification).
c. **Region Classification:** Used a grid system to classify brands’ locations as either rural, suburban, or rural based on the number of businesses within that grid.
d. Created new labels like **Is_Brand** (Chain vs. Independent) and **Urban_Rural** tags for classifying between urban and rural areas.
e. **Volatility Analysis:** Categorized businesses into groups and graphed the top 10 categories per state and analyzed their volatility based on an increasing number of reviews.

4. **Feature Engineering:**

Niche competition score: Calculated by grouping the original categories of the companies per city and standardizing average rating across companies within the same category in the same city.

$$\text{Niche} = \frac{\text{avg_rating} - \text{avg_competition_city}}{\text{std_competition_city}}$$

- **Broad competition score:** Calculated by grouping the categories of the companies into larger sub-categories per city and standardizing average rating across companies within the same sub category in the same city.

$$\text{Broad} = \frac{\text{avg_rating} - \text{avg_Broad_competition_city}}{\text{std_Broad_competition_city}}$$

- **City relative score:** Normalized average score of company to the relative average score in the city.

$$\text{City} = \frac{\text{avg_rating} - \text{avg_city}}{\text{std_city}}$$

- **Category relative score:** Normalized Category relative score to the business based on its specific category.

$$\text{Category} = \frac{\text{avg_rating} - \text{avg_category}}{\text{std_category}}$$

- **Opportunity score:** A score that measures how good a city is for a new type of company.

$$\text{Opportunity} = \frac{(5 - \text{avg_comp_city}) \times (\text{num_rev_city})}{\text{num_rev_cat_city}}$$

- **Likely Rating Range:** A 95% confidence interval for the average_review in stars per company.

$$\text{Range} = \left(avg_rev - 1.96 \frac{std_avg_rev}{\sqrt{num_rev}},\ avg_rev + 1.96 \frac{std_avg_rev}{\sqrt{num_rev}} \right)$$

5. **Storage:**
   - **Tools:** `CSV`, `GitHub`.
   - **Components:**
     i. **Flattening: Took the complex, nested JSON data and "flatten" it into a clean, simple table (CSV).**
     ii. **Version Control: Used `GitHub` to ensure every team member is using the same cleaning script and the same data versions, and compared multiple iterations to decide which model to move forward with.**

6. **Presentation:**

- **Tools:** `Matplotlib`, `Plotly`, `Tableau, Google Slides`
- **Components:**
  - **Exploratory Visualization:** Using `Plotly and Matplotlib` inside our notebooks to create "technical" charts for the team to verify the data.
  - **Executive Dashboards:** Using `Tableau` to create the Public interface of the project—interactive maps, sentiment gauges, and filters.

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4. Technical Stack

Category	Tool	Implementation Detail

Development	VS Code	Environment for writing and testing Python script and Jupiter notebook.
Analysis	Python	Data processing and statistical analysis performed using libraries such as pandas, Numpy, and NTLK (VADER Sentiment) for text-based sentiment analysis.
Visualization	Plotly, matplotlib	Used for interactive exploratory data analysis (EDA) and distribution visualizations to identify trends and patterns.
BI Reporting	Tableau	Designed and developed the final interactive dashboard to visualize performance summary, insight, and recommendation.
Version Control	GitHub	Managing code, documentation, and version tracking throughout the project lifecycle.
Data Ingestion	Glob	Used to programmatically grab all `.json.gz` files from the `/data` folder.
Statistical Analysis	Scikit-Learn	Used for `StandardScaler` to normalize scores across different states.

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5. Data Schema

Review Datasets

Column Name	Data Type	Description
user_id	String	Unique identifier for the user.
name	String	Name of reviewer.
time	Integer	Unix timestamp for when review was posted.
rating	Integer	1-5 star rating.
text	String	Review of business.
pics	Object	Picture attached to review (if any).
resp	Object	Response to review.
gmap_id	String	Unique identifier for business.

Metadata Datasets

Column Name	Data Type	Description
name	String	Name of business
address	String	Address of business
gmap_id	String	Unique identifier for business
description	String	Description of business
latitude	Float	Latitude of business location.
longitude	Float	Longitude of business location.
category	List	List of relevant business categories.
avg_rating	Float	Average rating of business.
num_of_reviews	Integer	Number of reviews of business.

price	String	$-$$$$ rating of relative price of business.
hours	List	Open hours of business.
MISC	Object	Additional business attributes (e.g. Accessibility, Service options)
state	String	Current operational status.
relative_results	Object	IDs for related/nearby businesses.
url	String	URL of business on Google Maps.

Sample review:


```
{
  'user_id': '106533466896145407182',
  'name': 'Amy VG',
  'time': 1568748357166,
  'rating': 5,
  'text': "I can't say I've ever been excited about a dentist visit before, but there's a first for everything! Loved my experience at Lush today. Every person in the office was
```

friendly and personable- plus the office itself is gorgeous! Great experience, I highly recommend!",
  'pics': [
  {
    'url':
['https://lh5.googleusercontent.com/p/AF1QipMBzN4BJV9YCObcw_ifNzFPm-u38hO3oimOA8Fb=w150-h150-k-no-p']
  },
  {
    'url':
['https://lh5.googleusercontent.com/p/AF1QipNS1PEXEvadfUlhRkRDJ09id
    Mxh3CveZGZYuTo5=w150-h150-k-no-p']
  }
  ],
  'resp': {
    'time': 1568770503975,
    'text': 'We love getting to meet new patients like yourself. Thanks for giving our office a chance to take care of your dental needs and thanks for the nice review!'
  },
  'gmap_id': '0x87ec2394c2cd9d2d:0xd1119cfbee0da6f3'
}

{
  'user_id': '101463350189962023774',
  'name': 'Jordan Adams',
  'time': 1627750414677,
  'rating': 5,
  'text': 'Cool place, great people, awesome dentist!',
  'pics': [
  {
    'url':
['https://lh5.googleusercontent.com/p/AF1QipNq2nZC5TH4_M7h5xRAd
    61hoTgvY1o9lozABguI=w150-h150-k-no-p']
  }
  ],
  'resp': {
    'time': 1628455067818,
    'text': 'Thank you for your five-star review! -Dr. Blake'
  },

```
  'gmap_id': '0x87ec2394c2cd9d2d:0xd1119cfbee0da6f3'
}
```

where

- user_id - ID of the reviewer
- name - name of the reviewer
- time - time of the review (unix time)
- rating - rating of the business
- text - text of the review
- pics - pictures of the review
- resp - business response to the review including unix time and text of the response
- gmap_id - ID of the business

Metadata

Sample metadata:


```
{
  'name': 'Walgreens Pharmacy',
  'address': 'Walgreens Pharmacy, 124 E North St, Kendallville,
IN 46755',
  'gmap_id': '0x881614ce7c13acbb:0x5c7b18bbf6ec4f7e',
  'description': 'Department of the Walgreens chain providing
prescription medications & other health-related items.',
  'latitude': 41.451859999999996,
  'longitude': -85.2666757,
  'category': ['Pharmacy'],
  'avg_rating': 4.2,
  'num_of_reviews': 5,
  'price': '$$',
  'hours': [['Thursday', '8AM-1:30PM'], ['Friday',
'8AM-1:30PM'], ['Saturday', '9AM-1:30PM'], ['Sunday',
'10AM-1:30PM'], ['Monday', '8AM-1:30PM'], ['Tuesday',
'8AM-1:30PM'], ['Wednesday', '8AM-1:30PM']],
  'MISC': {
    'Service options': ['Curbside pickup', 'Drive-through',
'In-store pickup', 'In-store shopping'],
    'Health & safety': ['Mask required', 'Staff wear masks',
'Staff get temperature checks'],
```

```
    'Accessibility': ['Wheelchair accessible entrance',
'Wheelchair accessible parking lot'],
    'Planning': ['Quick visit'],
    'Payments': ['Checks', 'Debit cards']
  },
  'state': 'Closes soon · 1:30PM · Reopens 2PM',
  'relative_results': ['0x881614cd49e4fa33:0x2d507c24ff4f1c74',
'0x8816145bf5141c89:0x535c1d605109f94b',
'0x881614cda24cc591:0xca426e3a9b826432',
'0x88162894d98b91ef:0xd139b34de70d3e03',
'0x881615400b5e57f9:0xc56d17dbe420a67f'],
  'url':
'https://www.google.com/maps/place//data=!4m2!3m1!1s0x881614ce7c
13acb
  b:0x5c7b18bbf6ec4f7e?authuser=-1&hl=en&gl=us'
}
```


where

- name - name of the business
- address - address of the business
- gmap_id - ID of the business
- description - description of the business
- latitude - latitude of the business
- longitude - longitude of the business
- category - category of the business
- avg_rating - average rating of the business
- num_of_reviews - number of reviews
- price - price of the business
- hours - open hours
- MISC - MISC information
- state - the current status of the business (e.g., permanently closed)
- relative_results - relative businesses recommended by Google
- url - URL of the business

United States Cities Database Example:

	city	city_ascii	state_id	state_name	county_fips	county_name	lat	lng	population	density	source	military
0	New York	New York	NY	New York	36081	Queens	40.6943	-73.9249	19268388	10943.7	shape	False
1	Los Angeles	Los Angeles	CA	California	6037	Los Angeles	34.1141	-118.4068	11984083	3165.7	shape	False
2	Chicago	Chicago	IL	Illinois	17031	Cook	41.8375	-87.6866	8609571	4590.3	shape	False
3	Miami	Miami	FL	Florida	12086	Miami-Dade	25.7840	-80.2101	6391670	4791.1	shape	False

Google Business Categories Dataset Example:

	industry	small_group	category_name
0	Agriculture & Natural Resources	Agricultural Services	agricultural
1	Agriculture & Natural Resources	Agricultural Services	agricultural association
2	Agriculture & Natural Resources	Agricultural Services	agricultural cooperative
3	Agriculture & Natural Resources	Agricultural Services	agricultural organization
4	Agriculture & Natural Resources	Agricultural Services	agricultural production
...	...	...	...
4269	Sports	Other Sports	table tennis facility
4270	Sports	Sports Facilities	american football field
4271	Sports	Sports Facilities	baseball field
4272	Sports	Sports Facilities	football field
4273	Sports	Sports Facilities	soccer field

6. Project Roadmap & Deliverables

- [ ] **Phase 1: Setup** - GitHub Repo initialized and datasets downloaded and converted to dataframes.
- [ ] **Phase 2: Early EDA**- Exploratory EDA in pandas via jupyter notebooks to identify questions and goals of the project
- [ ] **Phase 3: Picked Direction** - Split the project into two main goals: 1. Help customers choose the best company available by reevaluating ratings 2. Help recommend locations for companies.
- [ ] **Phase 4: Cleaning and EDA** - Searched data for outliers including geographic outliers, “Permanently closed” companies, null values, duplicate data values, and lack of text data. Cleaned certain values and handled outliers.
- [ ] **Phase 5: Feature engineering and Data transformation** - created z-scores across business categories as well as confidence intervals taking into account the number of reviews. Attempted sentiment analysis and classification of brands. Geo mapped differences between Rural, suburban, and Urban areas for further data analysis.
- [ ] **Phase 6: Dashboard creation** - familiarized ourselves with the Tableau software. Performed final data transformation before loading CSV formatted Data into Tableau for our respective states. Brainstormed and began basic interactive plots with Tableau.
- [ ] **Phase 7: PHD student meeting** - We presented our project progress and gathered student feedback to guide the final development phase. We decided to group businesses by their closest city to better reflect geographic review culture. After unsuccessful attempts to group categories using common words and NLP, we instead merged with the Google Common Category dataset to create broader category groupings for Tableau. We also chose to build multiple visualizations for both customer and business perspectives, and separated sentiment analysis for restaurants and non-restaurants due to differences in review language and sentiment patterns.
- [ ] **Phase 8: Final Feature engineering** - For the customer side implemented a niche competition score, broad competition score, relative city score, and relative category score. For the business side we created an opportunity score in order to help businesses locate.
- Pivoted our sentiment analysis approach by counting words in reviews rated **1–2 stars** and labeling them as “bad,” and counting words in **4–5 star reviews** and labeling them as “good.”
- [ ] **Phase 9: Finalize Dashboard:** Load final data sets with all features included for Texas, California, New York, Vermont, Virginia, and Florida to make interactive plots that achieve both goals from.
- [ ] **Phase 10: Finalize Project:** Finish design document, polish github and include a README, create presentation slides, create live demo for Dino Cage Presentation, and practice Dino Cage presentation as a group beforehand.

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7. Directory Structure

```
├── data/
│   ├── raw (metadata and reviews datasets)/          # Original, untouched data
│   └── processed
(https://drive.google.com/drive/folders/1qB_tD36dqhy8q3UVVJqxr3pASNrUNSBu)/
# Cleaned data ready
├── notebooks/         # .ipynb files for EDA and Tableau
│   ├──Brands_and_Reviews_Analysis/          # Analysis of Branding and Reviews Raw Datasets
│   ├──Generalized_Reviews_Analysis/          # Generalized Analysis of Reviews Raw Datasets
│   ├──Normalizing_Review_Culture/          # Score notebook for Tableau
│   ├──Volatility_analysis_refined/          # Volatility Analysis
│   └── rural_vs_suburban_vs_urban_refined/      # Region analysis and classification

├── outputs/          # Exported charts and PDF reports
│   └── Tableau Combined Dashboard/      # Combined Processed Data Into A Dashboard
├── README.md          # Project entry point
└── DESIGN_DOC.md       # This document
```